

Correct Answers, Uneven Grounding

The distributed-evidence gap in legal RAG

Answers remain similarly accurate. Complete source support falls when evidence spans pages.

-12.2 pp
page-grounding gap
95% CI -19.0 to -5.9

95 eligible questions within a 100-question benchmark / 30 PDFs / 590 pages

1 HYPOTHESIS

A model can answer from retrieved evidence yet emit an incomplete source set. Aggregate scores can hide whether this mismatch depends on evidence shape. [1,4,5]

RESEARCH QUESTION

How does distributed evidence affect the relationship between answer quality and source grounding in a legal RAG pipeline?

H1

Questions with multi-page gold evidence will have comparable answer scores but lower final page grounding. Uniform citation narrowing may widen that gap.

59

● localized

exactly one gold page

36

◆ distributed

multiple gold pages

Answer quality / mean benchmark score on a 0-1 scale.

Source grounding / page F-beta, beta = 2.5, weighting recall.

Evidence shape / 35 of 36 distributed questions also span documents; the effects cannot be separated.

2 METHODOLOGY

01

Page-first ingestion

30 public DIFC PDFs / 590 pages

02

Multi-view index

Page, section, clause, chunk, table

03

Hybrid retrieval

BM25 + Qwen3 Embedding

04

Fuse and rerank

RRF + Qwen3 Reranker

05

Typed answering

gpt-5.4-mini + validation

06

Page attribution

Supported pages + auditable trace

● Localized / one gold page ◆ Distributed / multiple gold pages

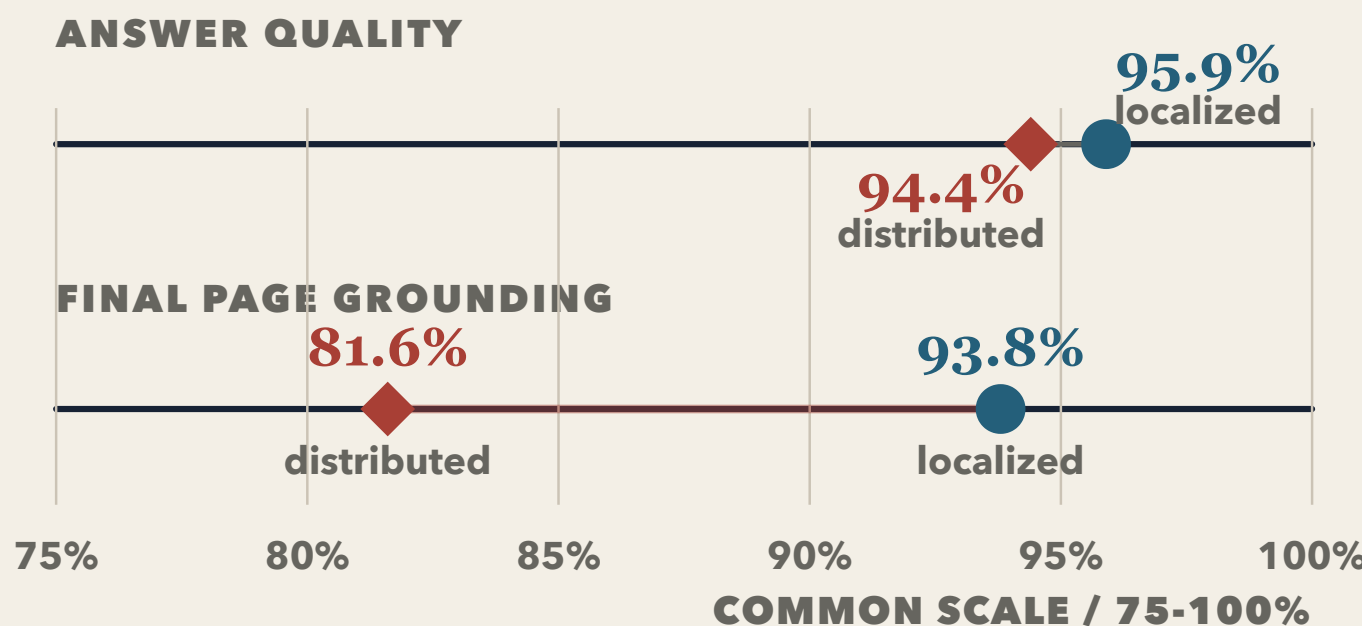
Explanatory flow only: the architecture makes evidence traceable; it does not prove which component caused the observed gap.

Page identity is preserved from ingestion to emitted citations.

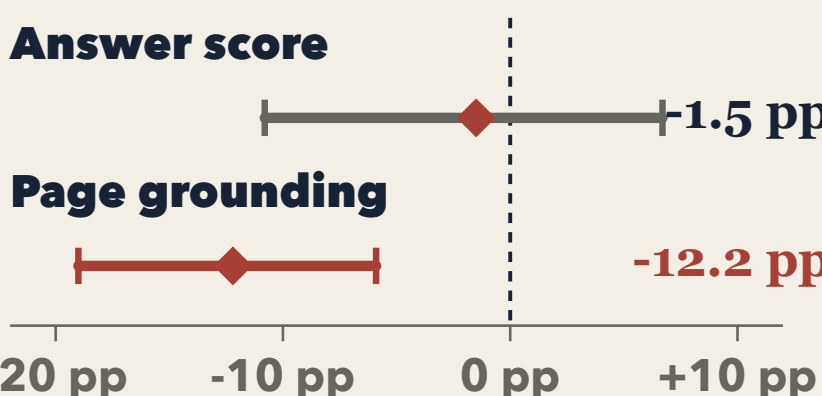
3 RESULTS

Answers stay level.
Grounding separates.

Means across the 95 questions with annotated gold pages.



DISTRIBUTED MINUS LOCALIZED



20,000 stratified bootstrap resamples. The answer interval crosses zero; the grounding interval does not.

PERFECT ANSWER / GROUNDING BELOW 0.8

● Localized 4 / 55 7.3%

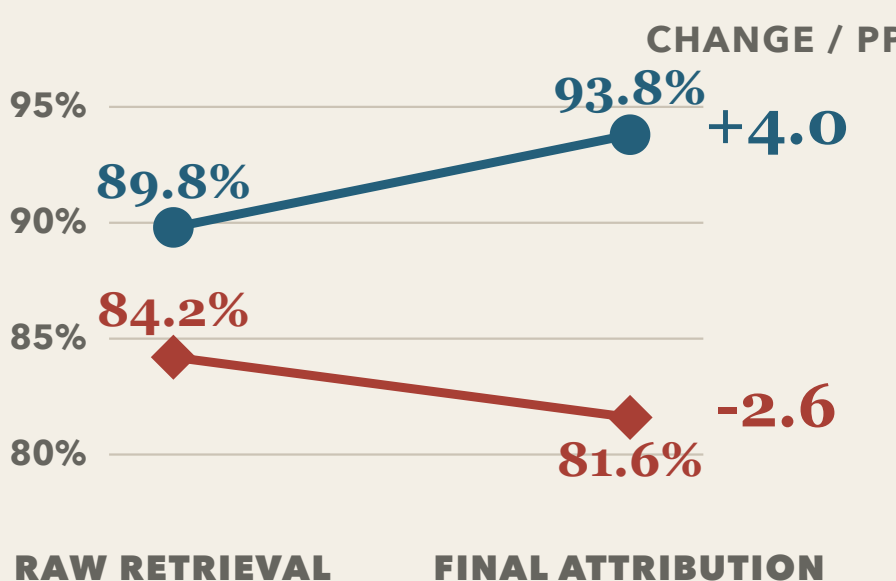
◆ Distributed 16 / 34 47.1%

Among answers scoring 1.0, incomplete page support is much more common for distributed evidence.

4 RESULTS / DIAGNOSTIC

Citation narrowing moves groups apart.

Page-level F-beta from the same run, before and after answer-conditioned attribution.



DECISION

Use evidence spread as a routing signal: narrow localized support; preserve and aggregate distributed support.

Gold evidence shape is unavailable at inference time. Use observable proxies: retrieval spread, cross-document dependencies, or claim-level support checks.

LIMITATIONS

One saved development run; retrospective hypothesis; observational groups; page and document spread are confounded; answer types and scoring methods are uneven; the grounding-below-0.8 threshold is descriptive; stage changes are not a causal ablation.

REPRODUCIBILITY

github.com/SteinVR/Agentic-RAG-Challenge

20,000 stratified bootstrap resamples / seed 20260903. Saved-run artifacts include **`solver_narrowing_true.yaml`**, **`benchmark_answer_scores.checkpoint.jsonl`**, and **`grounding/ledger.csv`**.

SELECTED REFERENCES

[1] Lewis et al., NeurIPS 2020 / RAG. [2] Karpukhin et al., EMNLP 2020 / dense retrieval. [3] Cormack et al., SIGIR 2009 / RRF. [4] Gao et al., EMNLP 2023 / citations. [5] Pipitone and Houir Alami, 2024 / LegalBench-RAG. [6] Zhang et al., 2025 / Qwen3 Embedding.